



How Data Scarcity Affects LLM Training for Low-Resource African Languages

Quantifying the web-data wall for 64 African languages

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MOTIVATION

LLMs scale with data, but the data is not equally distributed.

LLMs follow predictable scaling laws when training data is abundant.
Performance grows with parameters, tokens, and compute.

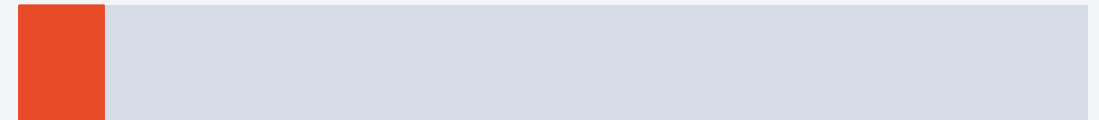
This breaks down for the 2,000+ languages of Africa.
Web corpora, filters, and tokenizers were not designed for them.

2,000+

Languages spoken across Africa

64African languages in AfroBench
evaluation suite**tens of pp**Performance gap vs. English on
IrokoBench

THE SCALING ASSUMPTION

more compute + more data = better models**English***trillions of tokens available***Yoruba, Hausa, Swahili, Zulu...***orders of magnitude smaller*

If raw data is input to scaling, what happens when there is barely any input?

The Scaling Promise — and Its African Exception

THE PROMISE

Scale solves everything

- LLMs follow smooth power laws — accuracy predicted from parameters, tokens, & FLOPs.
- Implicit assumption: more web text is always available.
- Train longer, scrape harder — performance follows.

THE EXCEPTION

The corpus is not infinite

- 2,000+ African languages. Hundreds of millions of speakers.
- AfroBench & IrokoBench: 10–30 pp gaps vs English.
- Web data may already be exhausted — scaling stops working.

RESEARCH QUESTION

**How much usable pre-training data
actually exists on the public web
for each African language?**

We move beyond raw byte counts to account for filtering, deduplication, and tokenizer inefficiency — yielding the first English-equivalent effective corpus estimate for 64 African languages.

Effective Corpus Framework — Three Stages

01 RAW WEB VOLUME

$$A_{raw,L}$$

Bytes addressable on the public web for language L (Common Crawl + GlotCC).

02 QUALITY FILTERING

$$A_{eff,L} = A_{raw,L} (1 - \alpha_L)$$

Apply filters:

1. Language ID
2. MinHash deduplication
3. Quality rules
4. Perplexity.

Measure noise rate α_L .

03 TOKENIZER FERTILITY

$$D_{eff,L} = A_{eff,L} / \varphi_L$$

Adjust for token tax φ_L

Tokenizers: Llama-3.1, Qwen-2.5, Gemma-2

Develop English-equivalent corpus.

Each language is reduced to a single, comparable English-equivalent number.

Step 1 — Estimating Raw Web Volume

SOURCE

Common Crawl 2023–2025 + GlotCC (1,000+ languages)

URLS

Find URLs where L is the dominant language

HOSTS

Aggregate over all unique registered domains

BYTES

Sum warc-total-length from CDX index

TOKENS

Convert at 1 GB $\approx$ 200M tokens (conservative)

FORMULA

$$A_{raw,L} = (\sum_{h \in H} \text{bytes}(h, L)) \cdot \frac{200 \text{ M tokens}}{1 \text{ GB}}$$

SAMPLING NOTES

$$n(N) = \begin{cases} N, & N < n_{\min} \\ n_{\min} + (n_{\max} - n_{\min}) \frac{\log N - \log n_{\min}}{\log N_{\max} - \log n_{\min}}, & N \geq n_{\min} \end{cases}$$

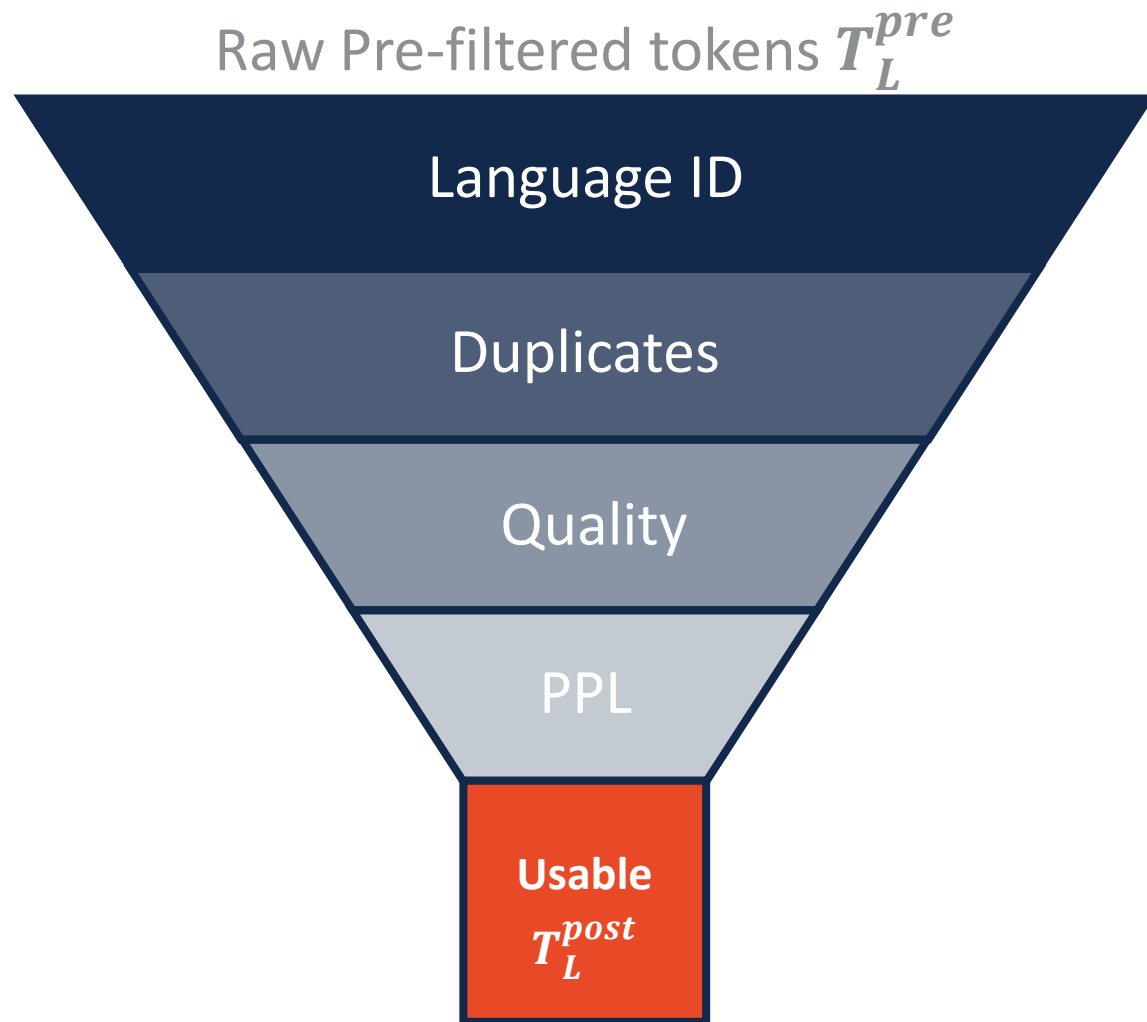
$$n_{\min} = 100,$$

$$n_{\max} = 1000,$$

$$N_{\max} = 120\text{M}$$

$$n = |h|$$

Step 2 — Quality Filtering Pipeline



NOISE RATE

$$\alpha_L = 1 - T_L^{post} / T_L^{pre}$$

Fraction of raw tokens removed by the filtering pipeline.

Larger $\alpha_L \rightarrow$ more low-quality web data for that language.

MAX EXTRACTABLE HIGH-QUALITY TOKENS

$$A_{eff,L} = A_{raw,L} (1 - \alpha_L)$$

The ceiling on clean, in-language tokens that any frontier-style pipeline could ever extract from the public web.

Step 3 — Tokenizer Fertility (the 'Token Tax')

WHY IT MATTERS

Morphologically rich languages need MORE subword tokens to express the same meaning as English — inflating compute cost.

EXAMPLE

Swahili	"nitakwenda"	→ 1 word
English	"I will go"	→ 3 words

Subword tokenizer breaks Swahili into many more pieces — same meaning, more tokens, more compute.

FERTILITY φ_L

$$\varphi_L^{(k)} = \frac{\left(T_L^{(k)} / W_L\right)}{\left(T_E^{(k)} / W_E\right)}$$

Averaged across three SOTA tokenizers:

Llama-3 · Qwen-2.5 · Gemma-2

$$\varphi > 1$$

compute penalty

$$\varphi < 1$$

no penalty

Sampling Setup — What We Filtered

64

African Languages

AfroBench coverage

10 GB

Raw HTML Sampled

≤160 MB / language

8,327

Documents Processed

post HTML stripping

FILTER STAGES APPLIED

1. Language ID (CLD2)
2. Near-duplicate removal
3. Quality scoring

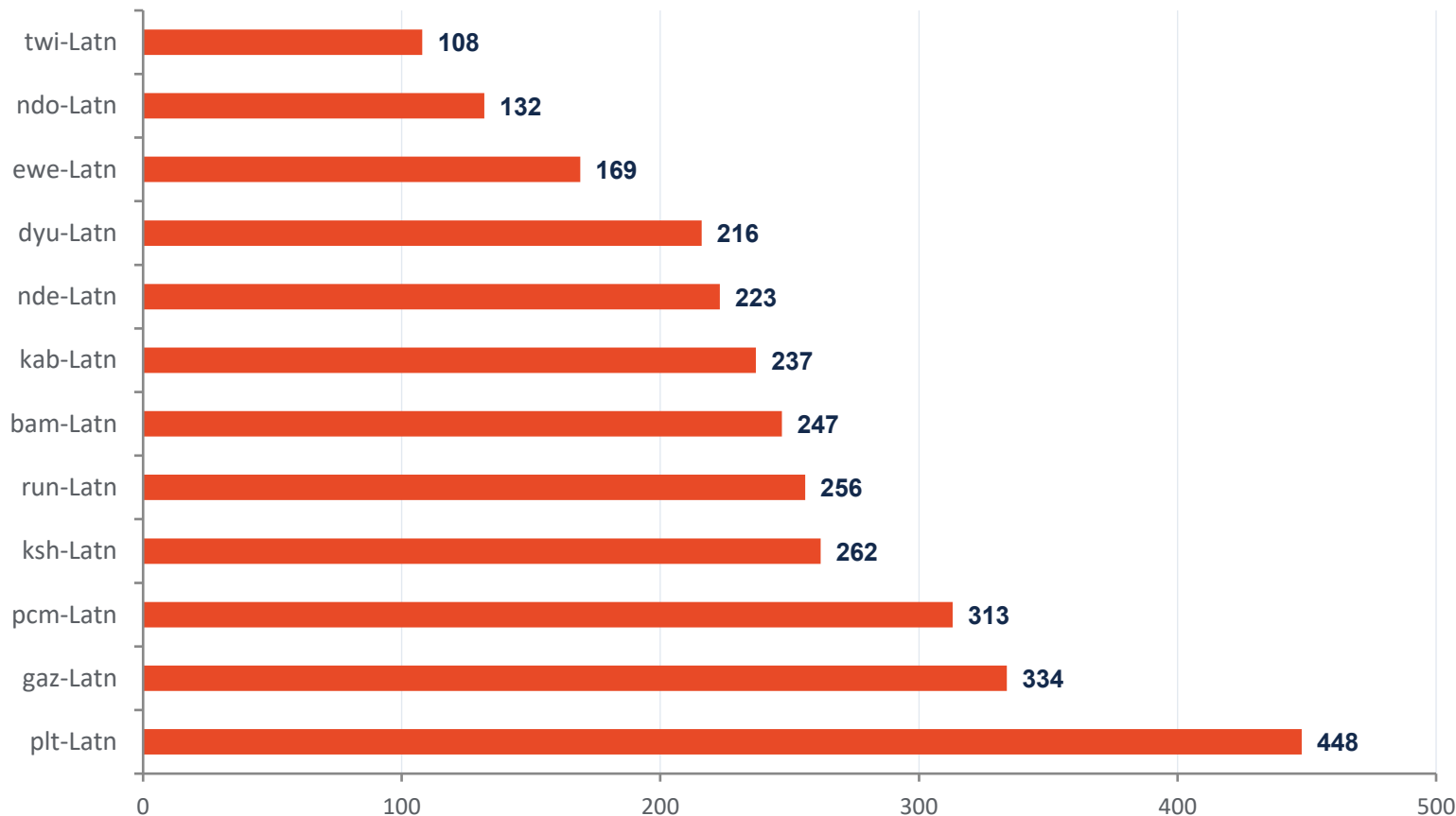
Confidence threshold; reject mismatched languages

MinHash, Jaccard similarity > 0.85

KenLM perplexity + length / charset rules

Result 1 — Language ID Nearly Erases Some Languages

Top 12 languages by absolute document loss (LangID stage)



KEY FINDING

Some languages lose nearly 100% at the very first filter.

- Plateau Malagasy: 448 / 448 docs lost
- Nigerian Pidgin: 313 / 313 lost
- *CLD2 fails to detect dominant language → discarded as 'wrong language.'*

Result 2 — Deduplication Trims a Smaller Slice

BEFORE DEDUP

8,327

documents

KEPT AFTER DEDUP

8,089

documents

REMOVED

238

near-duplicates

2.9%

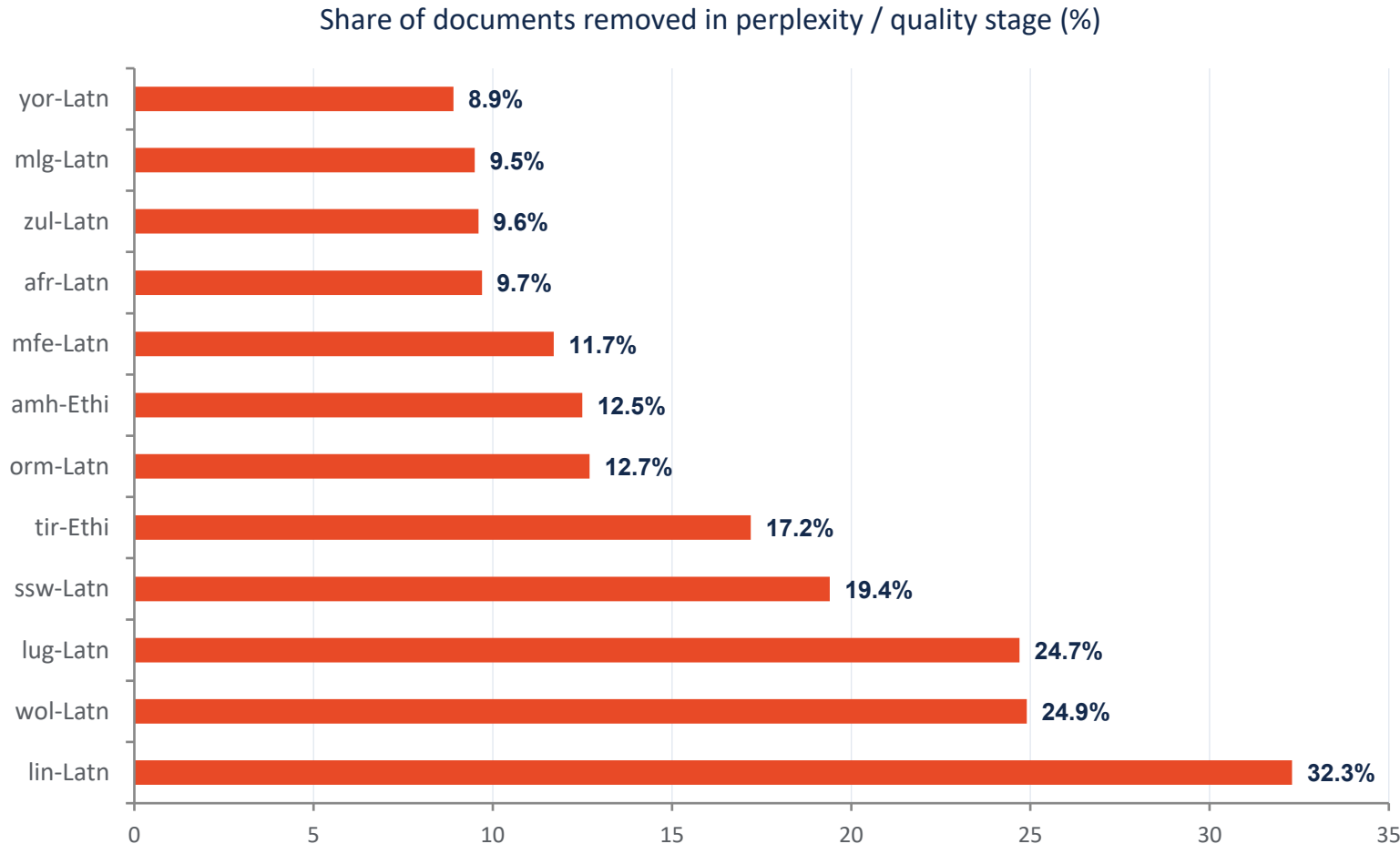
of documents

3.6%

of tokens (~0.7M)

After language matching, near-duplicates contribute relatively little additional removal — the long tail of unique-but-low-quality text is what hurts low-resource languages.

Result 3 — Quality Filtering Hits Languages Unequally



WHY?

Documents are too short for KenLM to score reliably.

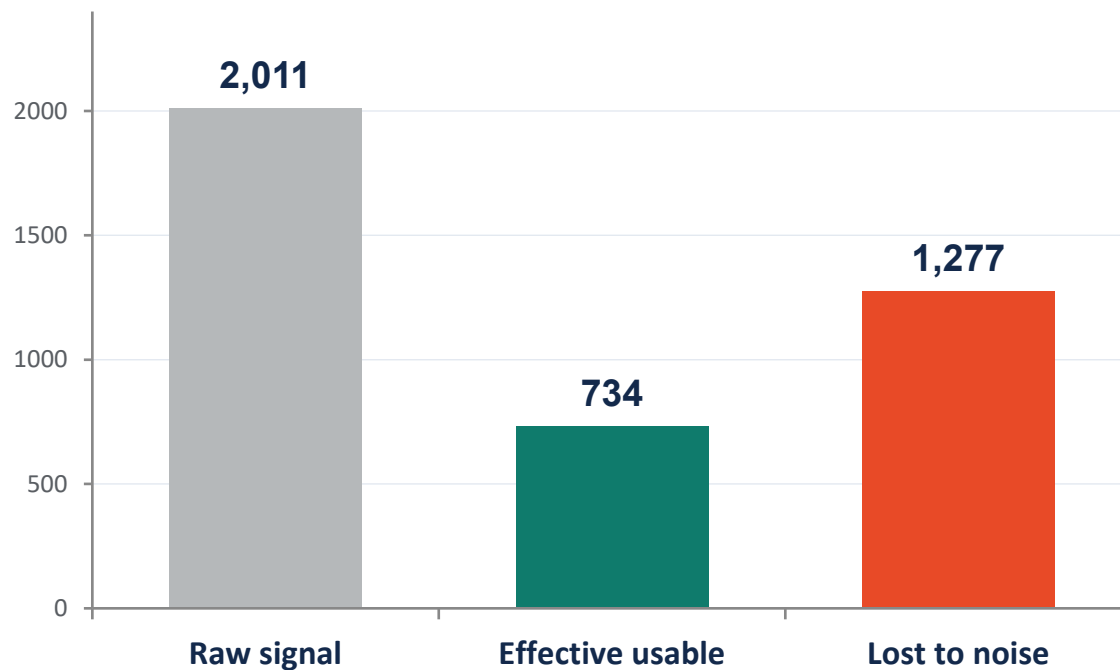
- Lingala loses 32.3% — to insufficient text length
- 0 documents fail the perplexity threshold
- *Length thresholds penalize languages with shorter native web pages.*

Result 4 — End-to-End Comparison: African Languages vs. English

DISCOUNTED AS NOISE (AFRICAN LANGS)

63.5%

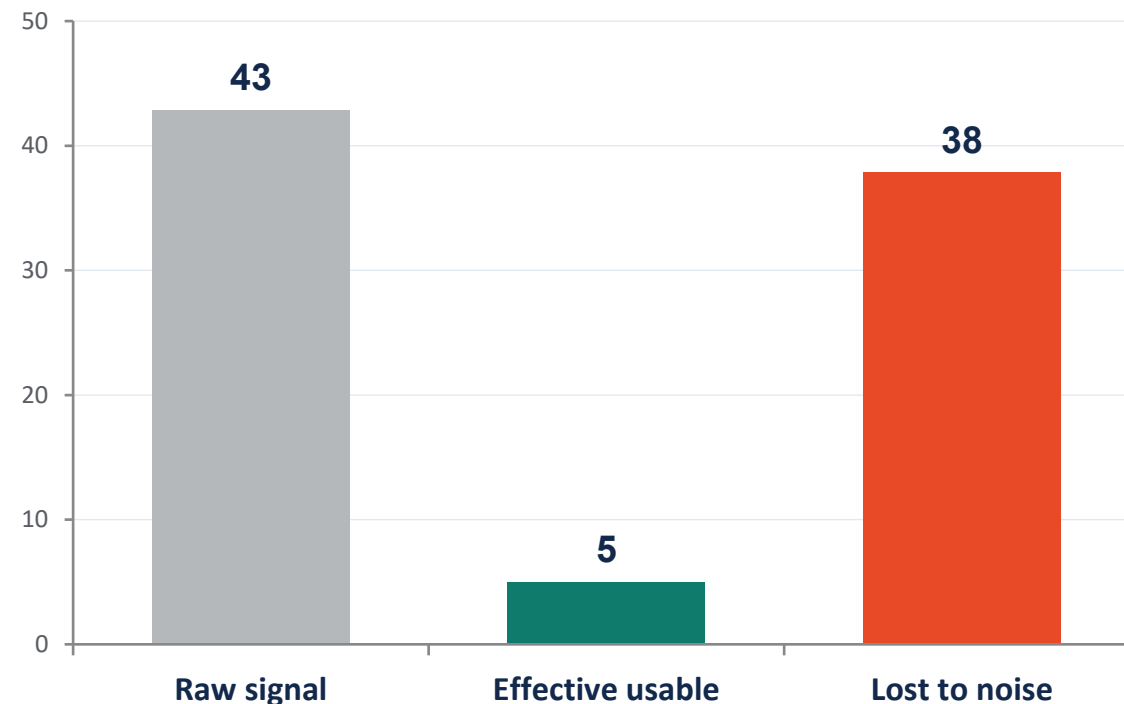
African Langs: Token budget across the full pipeline (millions)



DISCOUNTED AS NOISE (ENGLISH)

88.37%

English: Token budget across the full pipeline (trillions)



What This Means — Implications

**01**

Some languages are already web-exhausted

More compute will not help — the public corpus is the bottleneck, not the model.

**02**

Filtering pipelines silently disadvantage minority languages

LangID and length-based heuristics wipe out clean low-resource text the most.

**03**

Tokenizer fertility compounds the data gap into a compute gap

Same compute budget buys far less semantic content for morphologically rich languages.

**04**

Synthetic data and curation must replace 'scrape harder'

Community corpora, targeted OCR, and language-aware mixing become the path forward.

Limitations & Next Steps

LIMITATIONS

- Perplexity scoring may over-penalize genuinely in-language text for extremely low-resource languages.
- Whitespace tokenization underestimates morphological complexity in Bantu / Niger-Congo families.
- Bytes-to-tokens conversion is corpus-level, not document-level.
- Simulated pipeline may not match proprietary frontier stacks exactly.

NEXT STEPS

- Validate A_{eff} against curated corpora (OSCAR, mC4, Masakhane) per language.
- Align effective corpus with AfroBench / IrokoBench performance curves.
- Open-source the full pipeline upon paper acceptance.



Thank you

Questions & discussion

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